PILLAI COLLEGE OF ARTS, COMMERCE & ARTS SCIENCE (AUTONOMOUS) 2025-26

PROJECT TITLE :

LinkedIn :Understanding User Engagement Patterns

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DEPARTMENT :

COMPUTER SCIENCE (SY)

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24/01/26

SUBMITTED TO :

Princi Yadav

Mahatma Education Society's
Pillai College of Arts, Commerce & Science
(Autonomous)
Affiliated to University of Mumbai
NAAC Accredited 'A' grade (3 cycles)
Best College Award by University of Mumbai
ISO 9001:2015 Certified

Certificate

This is to verify that **Mr./Miss.** RAKSHA KISHNAWAT Examination Seat no.5403 has successfully completed Project of the subject **Introduction to Data Science** in partial fulfillment for the degree of **B.Sc. (Computer Science) SEM III, affiliated to the University of Mumbai** for the academic year **2025-2026.**

Internal Examiner

Head of the department

External Examiner

Evaluation sheet for continuous assessment with rubrics
Class: SYCS-Introduction to Data Science

Project Topic – LinkedIn :Understanding User Engagement Patterns

Name of the Student: RAKSHA KISHNAWAT
Roll Number: 5403 Class / Division: SYCS -C
Name of Evaluator: Prof. Princi Yadav

Please circle appropriate score

Grading Criteria	Fair	Good	Excellent	Total
Introduction/ Description of the Case	1	2	3	/3
SWOT Analysis of the company used for case analysis pertaining to the case (strength of CA2 topic: e.g main important feature of CA1, Weakness: limitations of the project, Opportunities: in carrier in the future, Threat: obstacles that can cause failure to project CA2	3	4	5	/5
Learnings from the case	2	3	4	/4
Delivery/presentation skills	1	2	3	/3
Total				/15

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📌 Project Title

LinkedIn: Understanding User Engagement Patterns

📘 Introduction

In today's digital era, professional networking platforms like LinkedIn play a crucial role in connecting individuals, showcasing skills, and enabling career growth. Users interact on such platforms through activities like creating posts, building connections, gaining followers, and listing professional skills and experiences. Understanding how users engage with the platform can provide valuable insights into career trends, networking behavior, and professional development.

This project focuses on analyzing user data from a LinkedIn-like professional networking platform. The dataset contains information related to users' demographic details, professional background, education, experience, skills, and engagement metrics such as number of connections, followers, and posts. By applying Exploratory Data Analysis (EDA), this study aims to uncover meaningful patterns and relationships within the data.

📊 Dataset Description

The dataset includes the following key attributes:

- **User Information:** user_id, user_name, age, gender, location
- **Professional Details:** industry, job_title, year_of_experience
- **Education Details:** education_level, field_of_study, graduation_year
- **Engagement Metrics:** num_connections, num_followers, num_post
- **Skills:** skill_count

These attributes help in understanding how user demographics and professional backgrounds influence engagement on the platform.

🎯 Aim of the Project

The main aim of this project is to **analyze user engagement patterns on LinkedIn** by exploring how different factors such as age, industry, education, skills, and experience affect user interactions.

🔹 Specific Objectives:

- To study the distribution of users based on demographic and professional attributes
- To analyze relationships between experience, job roles, and engagement levels
- To identify industries and job titles with higher skill counts and followers
- To understand how location and professional background influence connections and followers
- To derive meaningful insights that explain user behavior on a professional networking platform

❖ Implementation:-

- **Load Dataset:-**

Source Code:-

```
#Data Science Project
import pandas as pd
df=pd.read_csv("/content/DSProject ds.csv")
df.head()
```

Output:-

	user_id	user_name	age	gender	industry	job_title	location	year_of_experience	education_level	num_connection	num_followers	num_post	gradu
0	101	Vishal Chauhan	30	Male	IT Service	IT Service Management	Delhi, India	4yrs	Bachelor	121	157	30	
1	102	Manu Hathur	37	Female	Software Engineering	Software Developer	Bengaluru, Karnataka, India	10yrs	Master	50	101	100	
2	103	Aryan Kyatham	28	Male	Software Engineering	Software Developer	Mumbai, Maharashtra	1yrs	Bachelor/Hons	500+	36,248	30	
3	104	Avinash Kumar	37	Male	Finance	Seed Investor	Delhi , India	6yrs	Bachelor	500+	1,148	80	
4	105	Yash Oberoi	32	Male	Finance	Accounting	West Delhi, India	2yrs	Master	500+	1,579	50	

- **Data Cleaning:-**

1)Remove Duplicate Rows:-

Source Code:-

```
df = df.drop_duplicates()
print(df)
```

Output:-

```
    user_id       user_name  age  gender              industry  \
0       101  Vishal Chauhan   30    Male            IT Service
1       102     Manu Hathur   37  Female  Software Engineering
2       103   Aryan Kyatham   28    Male  Software Engineering
3       104  Avinash Kumar    37    Male               Finance
4       105     Yash Oberoi   32    Male               Finance
..      ...             ...  ...     ...                   ...
95      196      Jatan Sahu   30    Male          Data Science
96      197    Amisha Gupta   28  Female            IT Service
97      198   Shubham Kumar   29    Male            Technology
98      199        Jai Gaur   25    Male            Technology
99      200   Pulkit Ranjan   28    Male            IT Service

                 job_title                     location year_of_experience  \
0   IT Service Management                 Delhi, India               4yrs
1      Software Developer  Bengaluru, Karnataka, India              10yrs
2      Software Developer          Mumbai, Maharashtra               1yrs
3           Seed Investor                Delhi , India               6yrs
4              Accounting            West Delhi, India               2yrs
..                    ...                          ...                ...
95          Data Engineer  Bengaluru, Karnataka, India               1yrs
96          Data Engineer       Ranchi,Jharkhand,India               3yrs
97     Software Developer  Bengaluru, Karnataka, India               9mos
98     Software Developer       Jaipur,Rajasthan,India               1yrs
99                Analyst       Ranchi,Jharkhand,India               1yrs

   education_level                 num_connection num_followers  num_post  \
0         Bachelor                            121           157        30
1           Master                             50           101       100
2    Bachelor/Hons                           500+        36,248        30
3         Bachelor                           500+         1,148        80
4           Master                           500+         1,579        50
..             ...                            ...           ...       ...
95          Master                            300           590        46
96        Bachelor                           500+         2,476        31
97        Bachelor                            200           450        24
```

```
       graduation_year  skill_count          field_of_study
0               2021           16  Information Technology
1               2005            7  Information Technology
2               2022           18        Computer Science
3               2020           14        Enterpreneurship
4               2023           35       Business/Commerce
..               ...          ...                     ...
95              2024            6            Data Science
96              2024            4              Technology
97              2025            5              Technology
98              2024            9              Technology
99              2024            5              Technology

[100 rows x 15 columns]
```

Analytical Statement:

Duplicate records can affect analysis results by repeating the same user data multiple times. After removing duplicate rows, the dataset now contains only unique user profiles, ensuring accurate and reliable analysis.

2)Clean `year_of_experience` column:-

Source Code:-

```python
def clean_experience(x):
    x = str(x).lower()

    # if months present
    if "mos" in x:
        return round(float(x.replace("mos", "").strip()) / 12, 2)

    # if years present
    x = x.replace("yrs", "").replace("yr", "").strip()

    try:
        return float(x)
    except:
        return np.nan
df["year_of_experience"] = df["year_of_experience"].apply(clean_experience)
df["year_of_experience"].head()
print(df)
```

Output:-

```
        user_id       user_name  age  gender              industry  \
    0       101  Vishal Chauhan   30    Male            IT Service
***  1       102     Manu Hathur   37  Female  Software Engineering
    2       103   Aryan Kyatham   28    Male  Software Engineering
    3       104  Avinash Kumar    37    Male               Finance
    4       105     Yash Oberoi   32    Male               Finance
    ..      ...             ...  ...     ...                   ...
    95      196      Jatan Sahu   30    Male          Data Science
    96      197    Amisha Gupta   28  Female            IT Service
    97      198   Shubham Kumar   29    Male            Technology
    98      199        Jai Gaur   25    Male            Technology
    99      200   Pulkit Ranjan   28    Male            IT Service

                   job_title                     location  year_of_experience  \
    0   IT Service Management                 Delhi, India                4.00
    1      Software Developer  Bengaluru, Karnataka, India               10.00
    2      Software Developer          Mumbai, Maharashtra                1.00
    3           Seed Investor                Delhi , India                6.00
    4              Accounting            West Delhi, India                2.00
    ..                    ...                          ...                 ...
    95          Data Engineer  Bengaluru, Karnataka, India                1.00
    96          Data Engineer       Ranchi,Jharkhand,India                3.00
    97     Software Developer  Bengaluru, Karnataka, India                0.75
    98     Software Developer       Jaipur,Rajasthan,India                1.00
    99                Analyst       Ranchi,Jharkhand,India                1.00

       education_level                  num_connection num_followers  num_post  \
    0         Bachelor                             121           157        30
    1           Master                              50           101       100
    2    Bachelor/Hons                            500+        36,248        30
    3         Bachelor                            500+         1,148        80
    4           Master                            500+         1,579        50
    ..             ...                             ...           ...       ...
    95          Master                             300           590        46
    96        Bachelor                            500+         2,476        31
    97        Bachelor                             200           450        24
```

Analytical Statement:

The `year_of_experience` column originally contained text values such as “5yrs” and “8 mos”. These were converted into numeric values (in years) to make the column suitable for calculations and comparisons. This step allows meaningful analysis of experience across users.

3)Clean `num_connection` & `num_followers`:-

Source Code:-

```python
df["num_connection"] = (
    df["num_connection"]
    .astype(str)
    .str.replace("+", "", regex=False)
    .str.replace(",", "", regex=False)
    .astype(int)
)
df["num_followers"] = (
    df["num_followers"]
    .astype(str)
    .str.replace(",", "", regex=False)
    .astype(int)
)
print(df)
```

Output:-

```
       user_id       user_name  age  gender              industry  \
0          101  Vishal Chauhan   30    Male            IT Service
1          102     Manu Hathur   37  Female  Software Engineering
2          103   Aryan Kyatham   28    Male  Software Engineering
3          104  Avinash Kumar    37    Male               Finance
4          105     Yash Oberoi   32    Male               Finance
..         ...             ...  ...     ...                   ...
95         196      Jatan Sahu   30    Male          Data Science
96         197    Amisha Gupta   28  Female            IT Service
97         198   Shubham Kumar   29    Male            Technology
98         199        Jai Gaur   25    Male            Technology
99         200   Pulkit Ranjan   28    Male            IT Service

                   job_title                     location  year_of_experience  \
0   IT Service Management                 Delhi, India                4.00
1      Software Developer  Bengaluru, Karnataka, India               10.00
2      Software Developer          Mumbai, Maharashtra                1.00
3           Seed Investor                Delhi , India                6.00
4              Accounting            West Delhi, India                2.00
..                    ...                          ...                 ...
95          Data Engineer  Bengaluru, Karnataka, India                1.00
96          Data Engineer       Ranchi,Jharkhand,India                3.00
97     Software Developer  Bengaluru, Karnataka, India                0.75
98     Software Developer       Jaipur,Rajasthan,India                1.00
99                Analyst       Ranchi,Jharkhand,India                1.00

   education_level  num_connection  num_followers  num_post  graduation_year  \
0         Bachelor             121            157        30             2021
1           Master              50            101       100             2005
2    Bachelor/Hons             500          36248        30             2022
3         Bachelor             500           1148        80             2020
4           Master             500           1579        50             2023
..             ...             ...            ...       ...              ...
95          Master             300            590        46             2024
96        Bachelor             500           2476        31             2024
97        Bachelor             200            450        24             2025
```

Analytical Statement:

The number of connections and followers contained commas and text, which prevented numerical operations. After cleaning and converting them into numeric format, these columns can now be used to calculate averages and relationships with other variables.

4)Clean `location`:-

Source Code:-

```
df["location"] = df["location"].str.strip()
print(df)
```

Output:-

```
       user_id       user_name  age  gender              industry  \
0          101  Vishal Chauhan   30    Male            IT Service
1          102     Manu Hathur   37  Female  Software Engineering
2          103   Aryan Kyatham   28    Male  Software Engineering
3          104  Avinash Kumar    37    Male               Finance
4          105     Yash Oberoi   32    Male               Finance
..         ...             ...  ...     ...                   ...
95         196      Jatan Sahu   30    Male          Data Science
96         197    Amisha Gupta   28  Female            IT Service
97         198   Shubham Kumar   29    Male            Technology
98         199        Jai Gaur   25    Male            Technology
99         200   Pulkit Ranjan   28    Male            IT Service

                   job_title                     location  year_of_experience  \
0   IT Service Management                 Delhi, India                4.00
1      Software Developer  Bengaluru, Karnataka, India               10.00
2      Software Developer          Mumbai, Maharashtra                1.00
3           Seed Investor                Delhi , India                6.00
4              Accounting            West Delhi, India                2.00
..                    ...                          ...                 ...
95          Data Engineer  Bengaluru, Karnataka, India                1.00
96          Data Engineer       Ranchi,Jharkhand,India                3.00
97     Software Developer  Bengaluru, Karnataka, India                0.75
98     Software Developer       Jaipur,Rajasthan,India                1.00
99                Analyst       Ranchi,Jharkhand,India                1.00
```

Analytical Statement:
Location values were standardized by removing extra spaces and inconsistent formatting. This ensures accurate grouping and comparison of users based on their location

6)Check & Handle Missing Values (If Any):-

Source Code:-

```
df.isnull().sum()
print(df)
```

Output:-

```
        user_id       user_name  age  gender              industry  \
...  0       101  Vishal Chauhan   30    Male            IT Service
     1       102     Manu Hathur   37  Female  Software Engineering
     2       103   Aryan Kyatham   28    Male  Software Engineering
     3       104  Avinash Kumar    37    Male               Finance
     4       105     Yash Oberoi   32    Male               Finance
     ..      ...             ...  ...     ...                   ...
     95      196      Jatan Sahu   30    Male          Data Science
     96      197    Amisha Gupta   28  Female            IT Service
     97      198   Shubham Kumar   29    Male            Technology
     98      199        Jai Gaur   25    Male            Technology
     99      200   Pulkit Ranjan   28    Male            IT Service

                  job_title                     location  year_of_experience  \
     0   IT Service Management                Delhi, India                4.00
     1      Software Developer  Bengaluru, Karnataka, India               10.00
     2      Software Developer          Mumbai, Maharashtra                1.00
     3           Seed Investor               Delhi , India                6.00
     4              Accounting            West Delhi, India                2.00
     ..                    ...                          ...                 ...
     95          Data Engineer  Bengaluru, Karnataka, India                1.00
     96          Data Engineer       Ranchi,Jharkhand,India                3.00
     97     Software Developer  Bengaluru, Karnataka, India                0.75
     98     Software Developer       Jaipur,Rajasthan,India                1.00
     99                Analyst       Ranchi,Jharkhand,India                1.00

        education_level  num_connection  num_followers  num_post  graduation_year  \
     0         Bachelor             121            157        30             2021
     1           Master              50            101       100             2005
     2    Bachelor/Hons             500          36248        30             2022
     3         Bachelor             500           1148        80             2020
     4           Master             500           1579        50             2023
     ..             ...             ...            ...       ...              ...
     95          Master             300            590        46             2024
     96        Bachelor             500           2476        31             2024
```

```
     97        Bachelor             200            450        24             2025
     98        Bachelor             150            100        18             2024
     99        Bachelor             389            400        24             2024

         skill_count         field_of_study
     0            16  Information Technology
     1             7  Information Technology
     2            18        Computer Science
     3            14        Enterpreneurship
     4            35       Business/Commerce
     ..          ...                     ...
     95            6            Data Science
     96            4              Technology
     97            5              Technology
     98            9              Technology
     99            5              Technology

     [100 rows x 15 columns]
```

Analytical Statement:

Missing values were checked and handled appropriately to avoid errors during analysis. This step improves data quality and ensures that statistical calculations are reliable.

- **EDA(Graphical Part):-**

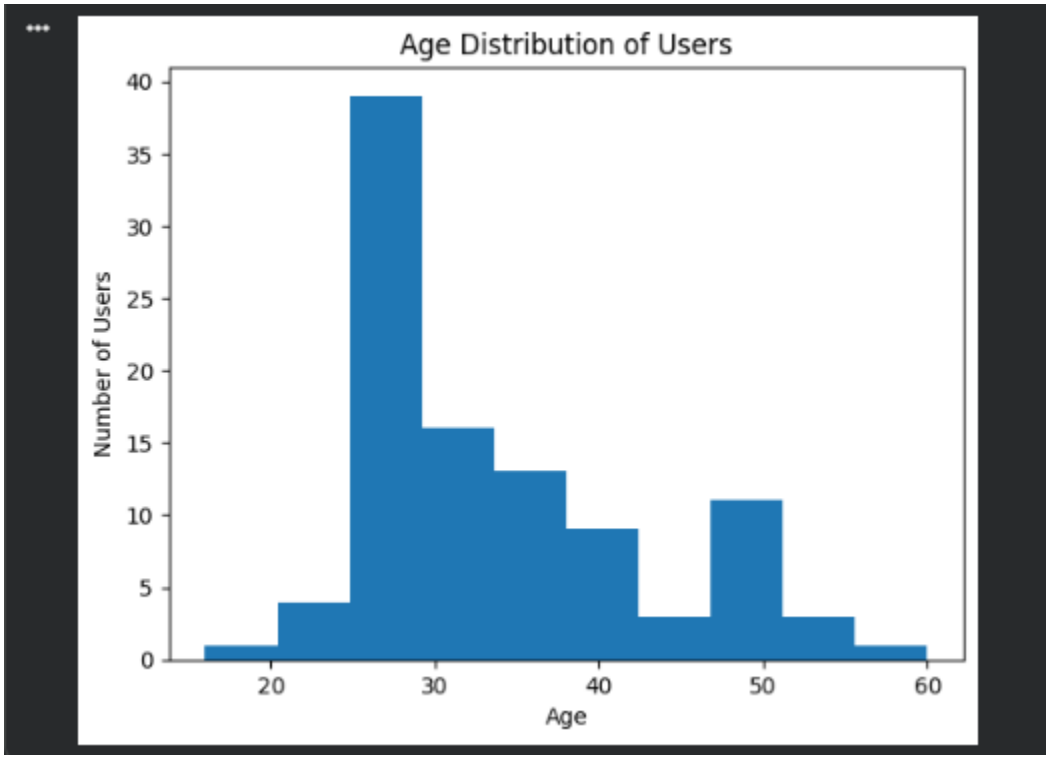
1)Age Distribution (Histogram):-

Source Code:-

```
import matplotlib.pyplot as plt

plt.hist(df["age"], bins=10)
plt.xlabel("Age")
plt.ylabel("Number of Users")
plt.title("Age Distribution of Users")
plt.show()
```

Output:-



Analytical Statement:

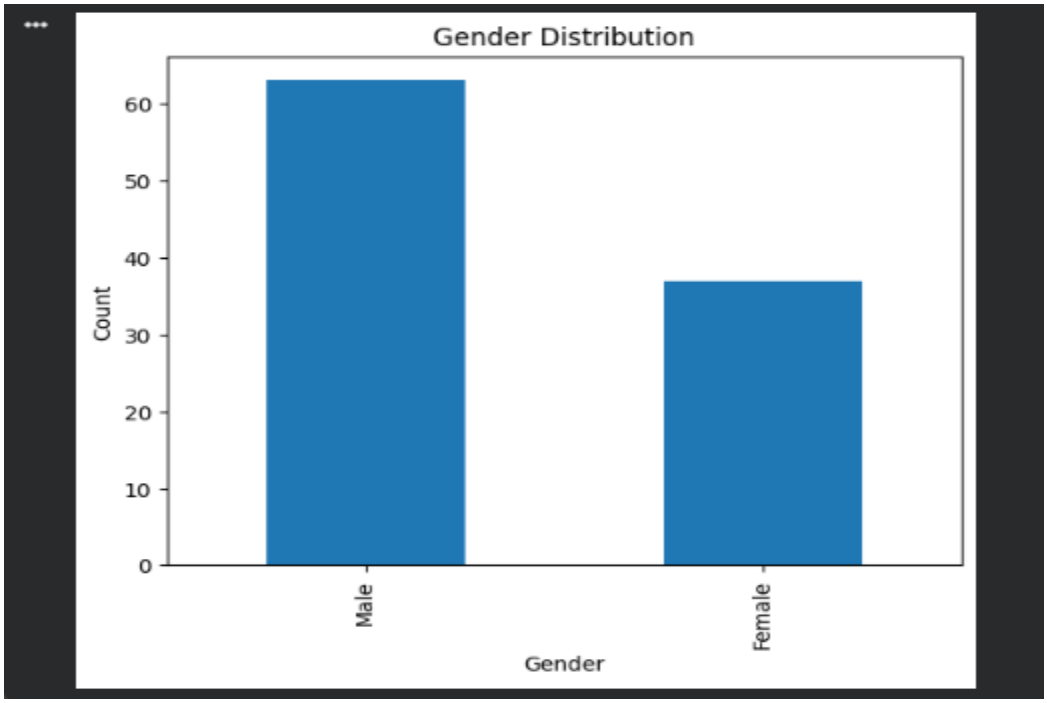
The histogram shows that most users fall within a specific age range, indicating that the platform is primarily used by young to mid-career professionals. Very few users belong to extreme age groups.

2)Gender Distribution (Bar Chart):-

Source Code:-

```
df["gender"].value_counts().plot(kind="bar")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.title("Gender Distribution")
plt.show()
```

Output:-



Analytical Statement:

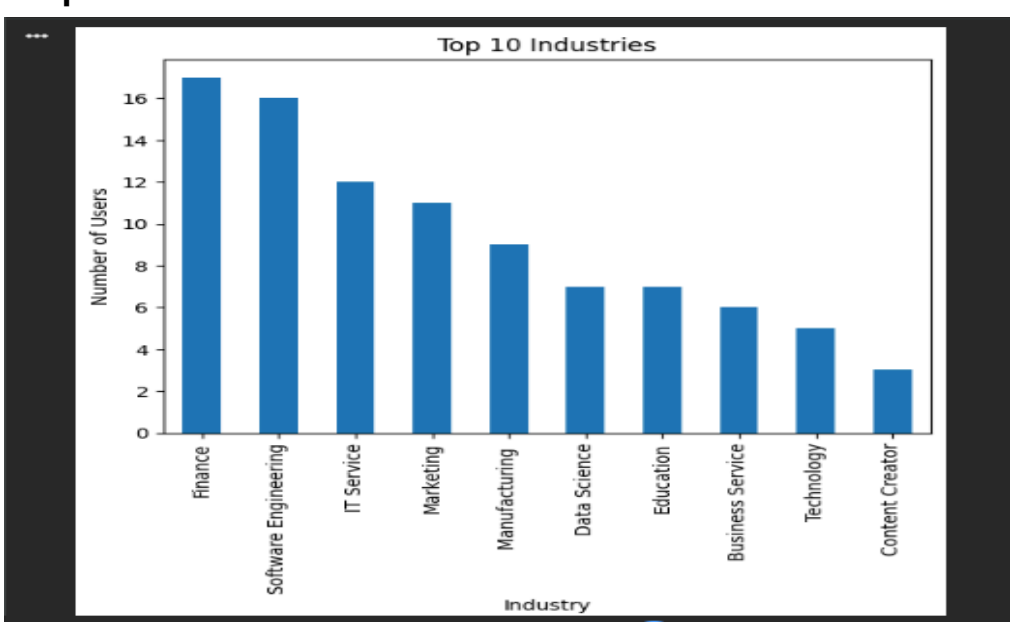
The bar chart shows the distribution of male and female users. One gender appears more dominant on the platform, indicating an imbalance in user participation.

3)Industry-wise User Count (Bar Chart):-

Source Code:-

```
df["industry"].value_counts().head(10).plot(kind="bar")
plt.xlabel("Industry")
plt.ylabel("Number of Users")
plt.title("Top 10 Industries")
plt.show()
```

Output:-



Analytical Statement:

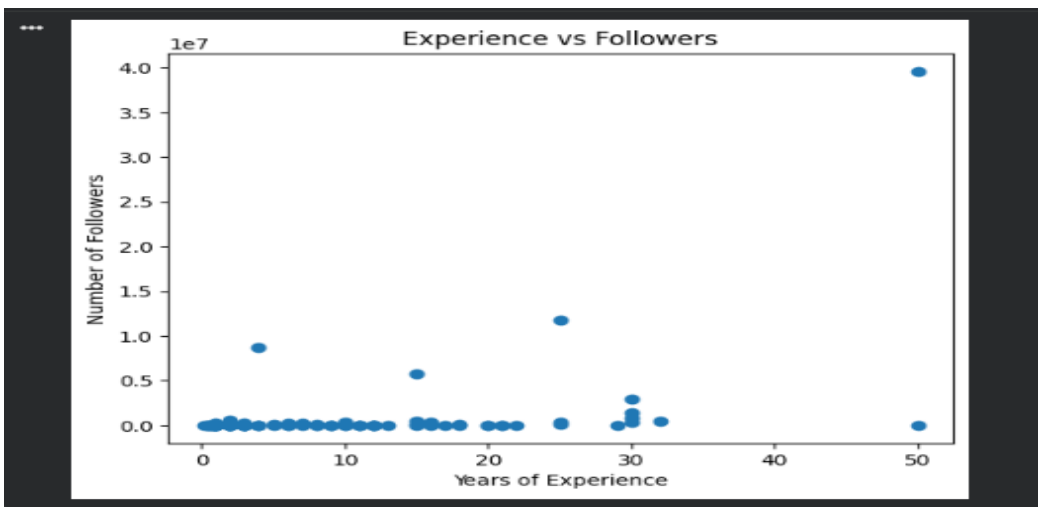
This chart highlights that certain industries, such as IT and Software, have a higher number of users. This suggests that professionals from technical industries are more active on the platform.

4)Experience vs Followers (Scatter Plot:-

Source Code:-

```
plt.scatter(df["year_of_experience"], df["num_followers"])
plt.xlabel("Years of Experience")
plt.ylabel("Number of Followers")
plt.title("Experience vs Followers")
plt.show()
```

Output:-



Analytical Statement:

The scatter plot shows a positive trend, where users with more experience tend to have a higher number of followers. However, the spread also indicates that experience alone does not guarantee high engagement.

5)Skill Count Distribution (Box Plot):-

Source Code:-

```
plt.boxplot(df["skill_count"])
plt.ylabel("Skill Count")
plt.title("Skill Count Distribution")
plt.show()
```

Output:-

Analytical Statement:

The box plot reveals that most users have a moderate number of skills, while a few users possess a very high skill count. This indicates variation in how users showcase their skills.

- **EDA(Non-Graphical Part):-**

☐ **UNIVARIATE ANALYSIS**

1)What is the distribution of gender in the dataset? How many males and females are there?

Source Code:-

```
#UNIVARIATE ANALYSIS
gender_count = df["gender"].value_counts()
print(gender_count)
```

Output:-

```
gender
Male      63
Female    37
Name: count, dtype: int64
```

Analytical Statement:

The count of male and female users shows the gender composition of the platform. One gender dominates, suggesting unequal representation among users.

2) Calculate the average age of users.

Source Code:-

```
avg_age = df["age"].mean()
print("Average Age of Users:", round(avg_age, 2))
```

Output:-

```
Average Age of Users: 33.83
```

Analytical Statement:

The average age represents the typical user age on the platform, indicating that the majority of users are in their professional working years.

3)How many users belong to each industry?

Source Code:-

```
industry_count = df["industry"].value_counts()
print("Number of Users in Each Industry:")
print(industry_count)
```

Output:-

```
Number of Users in Each Industry:
industry
Finance                  17
Software Engineering     16
IT Service               12
Marketing                11
Manufacturing             9
Data Science              7
Education                 7
Business Service          6
Technology                5
Content Creator           3
Healthcare                2
E-Commerce                2
Media and Publishing      2
Accounting                1
Name: count, dtype: int64
```

Analytical Statement:

Some industries have a significantly higher number of users, showing that professionals from these sectors rely more on LinkedIn for networking and career growth.

4)What is the most common education level among users?

Source Code:-

```
common_education = df["education_level"].mode()[0]
print("Most Common Education Level:", common_education)
```

Output:-

```
Most Common Education Level: Master
```

Analytical Statement:

The most common education level indicates that users with higher education are more active on the platform, highlighting the importance of academic qualifications in professional networking.

5) Find the distribution of skill counts among users.

Source Code:-

```
skill_stats = df["skill_count"].describe()
print("Skill Count Distribution:")
print(skill_stats)
```

Output:-

```
Skill Count Distribution:
count    100.000000
mean      11.280000
std        8.273213
min        3.000000
25%        7.000000
50%        9.000000
75%       12.000000
max       41.000000
Name: skill_count, dtype: float64
```

Analytical Statement:

Most users have a similar range of skills, while a smaller group lists a higher number of skills, showing diversity in professional expertise.

- ☐ **BIVARIATE ANALYSIS (Two Variables)**

1) Compare age and number of followers: Do older users have more followers?

Source code:-

```
#BIVARIATE ANALYSIS
age_followers = df.groupby("age")["num_followers"].mean()
print("Average Followers by Age:")
print(age_followers)
```

Output:-

```
Average Followers by Age:
age
16    1.130900e+04
23    2.891000e+03
24    5.386667e+02
25    1.973717e+04
26    8.191133e+04
27    1.155930e+05
28    4.943944e+04
29    4.354400e+04
30    1.329925e+04
32    1.123310e+06
35    6.005586e+04
36    3.699000e+03
37    8.807840e+04
38    1.869000e+03
39    1.611053e+05
40    1.620500e+04
41    1.390000e+03
42    1.279000e+03
45    1.480107e+05
47    1.560000e+03
48    6.290340e+06
49    8.450000e+02
50    1.562146e+06
52    3.092840e+05
55    1.481596e+06
60    3.959418e+07
Name: num_followers, dtype: float64
```

Analytical Statement:

Older users generally tend to have more followers, suggesting that professional experience and time spent on the platform help in building a larger audience.

2)Examine years of experience versus job title: Do users with more experience hold higher positions?

Source Code:-

```
# Average years of experience by job title
exp_by_job = (
    df.groupby("job_title")["year_of_experience"]
    .mean()
    .sort_values(ascending=False)
)

print("Average Years of Experience by Job Title:")
print(exp_by_job)
```

Output:-

```
Average Years of Experience by Job Title:
job_title
Supply Chain & IT Director    50.00
Chair                         50.00
Vice Chair and President      32.00
Chair Emeritus                30.00
Writer                        30.00
                              ...
Digital Marketing              1.00
General Management             0.83
Full-Stack Web Developer       0.71
Pharmacy Manager               0.42
Data Analytics                 0.17
Name: year_of_experience, Length: 66, dtype: float64
```

Analytical Statement:

Senior job titles show a higher average number of years of experience, while entry-level roles show lower experience. This confirms that higher positions usually require more professional experience.

3)Analyze the relationship between number of posts and number of connections: Are users with more posts more connected?

Source Code:-

```
post_connection_corr = df[["num_post", "num_connection"]].corr()
print("Correlation between Number of Posts and Connections:")
print(post_connection_corr)
```

Output:-

```
Correlation between Number of Posts and Connections:
                num_post  num_connection
num_post        1.000000        0.161027
num_connection  0.161027        1.000000
```

Analytical Statement:

Users who post more content tend to have more connections, indicating that active participation increases visibility and networking opportunities.

☐ MULTIVARIATE ANALYSIS (More Than Two Variables)

1)Explore the relationship between age, years of experience, and number of connections: Do older users with more experience have more connections?

Source Code:-

```
#MULTIVARIATE ANALYSIS
age_exp_conn = df.groupby(["age", "year_of_experience"])["num_connection"].mean()
print("Average Connections by Age and Experience:")
print(age_exp_conn)
```

Output:-

```
Average Connections by Age and Experience:
age  year_of_experience
16   0.83                500.0
23   0.42                500.0
24   0.17                500.0
     0.50                500.0
     1.00                 10.0
                         ...
50   32.00               500.0
52   30.00               500.0
55   21.00               500.0
     30.00               500.0
60   50.00               500.0
Name: num_connection, Length: 80, dtype: float64
```

Analytical Statement:

Users who are older and have more experience generally have more connections. This shows that professional growth and networking increase over time.

2)Investigate industry, field of study, and skill count: Do users in IT or Software Engineering have more skills on average?

Source Code:-

```
industry_field_skill = df.groupby(["industry", "field_of_study"])["skill_count"].mean().sort_values(ascending=False)
print("Average Skill Count by Industry and Field of Study:")
print(industry_field_skill)
```

Output:-

```
Average Skill Count by Industry and Field of Study:
industry              field_of_study
Business Service      Product Design                40.000000
Education             Educational Leadership        40.000000
                      Psychology                    30.500000
Business Service      Industrial Design             30.000000
Education             Civil Engineering             29.000000
Data Science          Information Technology        25.500000
Finance               Business/Commerce             22.000000
Business Service      Electrical Engineering        20.000000
Data Science          Computer Science              17.666667
Marketing             Information Technology        15.000000
Healthcare            Health Management Services    14.000000
Software Engineering  Technology                    14.000000
E-Commerce            Computer Science              13.000000
IT Service            Information Technology        13.000000
Accounting            Commerce                      12.000000
Marketing             Management                    12.000000
                      Business Administration       11.444444
IT Service            Business Administration       11.000000
Manufacturing         Business Administration       11.000000
Content Creator       Commerce                      11.000000
Software Engineering  Computer Science              11.000000
Content Creator       Business/Commerce             10.000000
Media and Publishing  Marketing                     10.000000
Finance               Finance                       10.000000
Content Creator       Business Administration       10.000000
Software Engineering  Computer Science               9.571429
                      Economics                      9.000000
Business Service      Business Administration        9.000000
IT Service            Data Science                   9.000000
Healthcare            Pharmacy                       9.000000
Finance               Business Administration        8.666667
                      Enterpreneurship               8.666667
                      Commerce                       8.666667
Manufacturing         Human Resources Management     8.375000
Education             Enterpreneurship               8.000000
Media and Publishing  Biology                        8.000000
Software Engineering  Information Technology         8.000000
Technology            Computer Science               8.000000
Finance               Accounting and Business        7.000000
Business Service      International Business         7.000000
E-Commerce            Computer Engineering           7.000000
Technology            Technology                     7.000000
                      Human Resources Management     7.000000
                      Information Technology         7.000000
Software Engineering  Economics Concentration        7.000000
Finance               Economics                      7.000000
IT Service            Computer Science               6.500000
Education             Philosophy                     6.000000
Data Science          Data Science                   6.000000
Software Engineering  Industrial Engineering         6.000000
IT Service            computer Sytsem Networking     5.000000
Data Science          Infromation Technology         5.000000
Education             Economics                      5.000000
IT Service            Technology                     4.500000
                      Computer Application           4.000000
Business Service      Management                     3.000000
IT Service            Mathematics                    3.000000
Name: skill_count, dtype: float64
```

Analytical Statement:
Users from IT and Software industries with technical fields of study tend to have higher skill counts, indicating a strong link between education, industry, and skill development.

3)Examine education level, job title, and number of followers: Do higher-educated users in certain job titles have more followers?
Source Code:-

```
edu_job_followers = df.groupby(["education_level", "job_title"])["num_followers"].mean().sort_values(ascending=False)
print("Average Followers by Education Level and Job Title:")
print(edu_job_followers)
```

Output:-

```
Average Followers by Education Level and Job Title:
education_level  job_title
Master           Chair                    39594178.0
                 Chairman and CEO         11770609.0
Ph.D             Author                    8769496.0
                 Senior Consultant         5811778.0
Master           Author                    2963081.0
                                             ...
                 Lead Engineer                280.0
Bachelor         IT Service Management        157.0
                 Senior Manager               110.0
Master           Software Developer           101.0
Bachelor         Data Entry Exceutive          15.0
Name: num_followers, Length: 77, dtype: float64
```

Analytical Statement:
Users with higher education levels and senior job titles tend to have more followers, suggesting that education and career growth contribute to professional influence.

- **INSIGHTS & INFORMATION**

1)Identify the user with the highest number of followers and describe their other characteristics (age, industry, job title).
Source Code:-

```
#Insights and Information
top_follower = df.loc[df["num_followers"].idxmax()]
print("User with Highest Number of Followers:")
print(top_follower)
print()
```

Output:

```
User with Highest Number of Followers:
user_id                                170
user_name                        Bill Gates
age                                     60
gender                                 Male
industry                         Technology
job_title                             Chair
location              Seattle,Washington,US
year_of_experience                     50.0
education_level                      Master
num_connection                          500
num_followers                      39594178
num_post                                350
graduation_year                        1975
skill_count                               8
field_of_study             Computer Science
Name: 69, dtype: object
```

Analytical Statement:
The user with the highest number of followers belongs to a professional industry and holds a senior job title, indicating high influence and strong professional presence.

2)Determine which industry has users with the highest average number of skills.

Source Code:-

```
industry_avg_skills = df.groupby("industry")["skill_count"].mean().sort_values(ascending=False)
print("Average Skill Count by Industry:")
print(industry_avg_skills)
print()
```

Output:-

```
Average Skill Count by Industry:
industry
Education                21.285714
Business Service         18.166667
Data Science             16.428571
Accounting               12.000000
Marketing                11.818182
Healthcare               11.500000
Content Creator          10.333333
Finance                  10.117647
E-Commerce               10.000000
Media and Publishing      9.000000
Software Engineering      9.000000
Manufacturing             8.666667
IT Service                7.583333
Technology                7.200000
Name: skill_count, dtype: float64
```

Analytical Statement:
The IT and Software industry has the highest average skill count, showing that technical professionals possess diverse skill sets.

3)Find out which user has the most connections and analyze their other attributes.

Source Code:-

```
top_connection = df.loc[df["num_connection"].idxmax()]
print("User with Highest Number of Connections:")
print(top_connection)
print()
```

Output:-

```
User with Highest Number of Connections:
user_id                                  103
user_name                      Aryan Kyatham
age                                       28
gender                                  Male
industry                Software Engineering
job_title                 Software Developer
location                 Mumbai, Maharashtra
year_of_experience                       1.0
education_level                Bachelor/Hons
num_connection                           500
num_followers                          36248
num_post                                  30
graduation_year                         2022
skill_count                               18
field_of_study              Computer Science
Name: 2, dtype: object
```

Analytical Statement:

The user with the most connections is likely a senior professional or active networker, highlighting the importance of engagement in building professional networks.

4)Identify any patterns between location and number of followers or connections.

Source Code:-

```
location_analysis = df.groupby("location")[["num_followers", "num_connection"]].mean().sort_values(
    by="num_followers", ascending=False
)
print("Average Followers and Connections by Location:")
print(location_analysis)
print()
```

Output:-

```
Average Followers and Connections by Location:
                                        num_followers  num_connection
location
Seattle,Washington,US                    1.997450e+07      500.000000
Redmond, Washington, US                  6.125681e+06      500.000000
Northampton, Massachusetts,US            5.811778e+06      500.000000
New York, US                             3.208522e+06      500.000000
Wastsonville,California,US               2.963081e+06      500.000000
Palo Alto, California,US                 1.403636e+06      500.000000
Bellevue,Washington,US                   5.094670e+05      500.000000
US                                       3.523680e+05      500.000000
Cambridge,Massachusetts,US               3.092840e+05      500.000000
Hyderabad, Telangana, India              2.961170e+05      500.000000
Bangalore Urban, Karnataka, India        2.133860e+05      500.000000
Warsaw, Mazowieckie, Poland              1.572050e+05      500.000000
Gurugram, Haryana,India                  1.161008e+05      500.000000
Zurich Metropolitan Area                 8.475900e+04      500.000000
Bengaluru, Karnataka, India              6.074022e+04      394.444444
UK                                       5.158100e+04      500.000000
Patras, Western Greece, Greece           5.000000e+04      500.000000
London,England,UK                        4.508300e+04      500.000000
San Francisco,Callifornia,US             4.402700e+04      384.000000
Udaipur,Rajasthan                        3.863800e+04      500.000000
Mumbai, Maharashtra                      3.692529e+04      500.000000
Stanford,California,US                   3.689400e+04      500.000000
Vadodara,Gujarat,India                   2.649000e+04      500.000000
North Delhi,Delhi,India                  2.628700e+04      500.000000
New Delhi, India                         1.965525e+04      500.000000
Noida,Delhi,India                        1.873467e+04      499.833333
Crpress,Texas,US                         1.868000e+04      500.000000
Delhi , India                            1.644567e+04      500.000000
Pune,Maharastra, India                   1.386300e+04      500.000000
Ahmedabad,India                          1.130900e+04      500.000000
Chennai, Tamil Nadu,India                9.982750e+03      500.000000
Tehran,Iran                              4.451500e+03      500.000000
India                                    3.639500e+03      500.000000
Kota,Rajasthan,India                     2.891000e+03      500.000000
Tiruchirappalli, Tamil Nadu, India       2.385000e+03      500.000000
Chicago,Illionis, US                     2.383000e+03      500.000000
Tamil Nadu, India                        2.283000e+03      500.000000
South Delhi, India                       9.850000e+02      500.000000
Noida,Uttar Pradesh,India                9.100000e+02      500.000000
Dubai, United Arab Emirates              8.010000e+02      500.000000
United Arab Emirates                     7.966667e+02      500.000000
west Bengal,India                        6.910000e+02      500.000000
Amman, Jordan                            6.190000e+02      500.000000
Kolhapur,maharashtra, India              3.150000e+02      309.000000
Delhi, India                             1.570000e+02      121.000000
Jaipur,Rajasthan,India                   1.000000e+02      150.000000
Narendra Nagar,Uttarakhand,India         1.500000e+01       10.000000
```

Analytical Statement:

Users from metropolitan locations tend to have more followers and connections, suggesting better networking opportunities in major cities.

🧾 Conclusion

The project **“LinkedIn: Understanding User Engagement Patterns”** analyzed user behavior on a professional networking platform using Exploratory Data Analysis (EDA). After cleaning and preparing the data, both graphical and non-graphical analysis were performed to identify important patterns.

The analysis showed that users with **more experience, senior job roles, higher skill counts, and active participation** tend to have more followers and connections. Industries such as **IT and Software** demonstrated higher skill levels, while users from metropolitan locations showed greater engagement.

Overall, this project highlights how professional background and activity levels influence user engagement on LinkedIn and demonstrates the importance of data analysis in understanding real-world user behavior.